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Opaque allomorph selection in Japanese and Harmonic Serialism: A reply to Kurisu (2012)

1 Introduction

Cross-linguistically, medial consonant clusters simplify by deleting or assimilating the first consonant /VtkV/ $\rightarrow$ [VkV] $\sim$ [VkkV], never the second *[VtV] (Wilson 2000; Steriade 2001). Parallel OT fails to capture this asymmetry, as the output consonant appears in the onset regardless of its position in the input. McCarthy (2007b, 2008) proposes a solution to this problem within Harmonic Serialism (HS), by postulating that deletion of the onset—i.e. the second consonant—involves a step which is not harmonically improving. The prediction that onset consonants never delete has been recognized as one of the crucial arguments for HS compared to parallel OT.

Kurisu (2012) challenges this generalization by bringing forth data from Japanese where onsets, not codas, appear to be deleted, presenting a problem for a two-step analysis in HS (§ 2). This squib takes a second look at the Japanese data by considering the full verbal paradigm (§ 3). The evidence instead suggests that the Japanese verbal paradigm pattern involves allomorph selection, not pure phonological deletion. Next, we show that allomorph selection opaquely interacts with another process. This situation cannot be formalized in parallel OT, but it can be captured in HS (§ 4). We then extend this finding to another case of opaque allomorphy found in Polish (§ 5), showing that the Japanese case is not isolated. Thus, Japanese does not present a counterexample to the generalization that onsets never delete in consonant cluster simplification, and in fact provides further support for HS.

2 Japanese verb suffixation and deletion

Japanese verb suffixes sometimes surface with an initial coronal (1). The generalization is that vowel-final stems are followed by a coronal in the suffix, whereas consonant-final stems are followed by vowel-initial suffixes.

(1) Japanese verbs (Kurisu 2012:311)

		/tob/ 'fly'	/ne/ 'sleep'
Infinitive	/-ru/	tob-u	ne-ru
Subjunctive	/-reba/	tob-eba	ne-reba
Causative	/-sase/	tob-ase	ne-sase
Volitional	/-joo/	tob-oo	ne-joo

Kurisu (2012) argues that underlying suffix-initial coronals are deleted when preceded by another consonant.[1] This deletion occurs because of the phonotactic restriction of Japanese: codas cannot have their own place specification. In OT, coronal deletion is driven by CODACOND(ITION) (Itô 1986/1988, 1989; Itô and Mester 1998, 2003; Goldsmith 1990). Furthermore, root segments are more faithful than suffix segments (MAX$_{\text{Root}}$ $\gg$ MAX$_{\text{Affix}}$; McCarthy and Prince 1995; Beckman 1998). In parallel OT, CODACOND can be satisfied by deletion of the second consonant, which would have been the onset in the output (2).

(2) Kurisu's parallel OT analysis

/tob-ru/	CODACOND	MAX$_{\text{Root}}$	MAX$_{\text{Affix}}$
a. ☞ to.bu			*
b. to.ru		*!	
c. tob.ru	*!		

As Kurisu (2012) points out, this analysis cannot be implemented in HS. HS is a variant of OT that combines constraint ranking with serial derivations (McCarthy 2010a,b, to appear). Gen in HS generates only those candidates that differ from the input by one single operation. The winning candidate is then fed back to Gen as a new input for another round of evaluation. This loop is then repeated until the fully faithful parse of the latest input wins.

Deletion has been considered a two-step process in HS: place features are removed first, followed by segment deletion (McCarthy 2007b, 2008). Thus at step 1, no segment can be entirely deleted, only debuccalized (3). The resulting segment is a consonant without a place feature, henceforth marked as “H”. The problem that the Japanese data in (1) present to the HS analysis is that the placeless onset candidate (a) should win at this step, but it is harmonically bounded by the placeless coda candidate (b) and the faithful candidate (c).

(3) Step 1: Placeless onset is harmonically bounded

/tob-ru/	CODACOND	MAX_{Root}	$\text{MAX}_{\text{Affix}}$	HAVEPLACE	MAX(PLACE)
a. ☹ tob.Hu	*!			*	*
b. ☞ toH.ru				*	*
c. tob.ru	*!				

Kurisu (2012) thus concludes that this paradox can only be resolved if deletion is a possible single-step operation, *contra* McCarthy (2007b, 2008). However, if deletion is a possible single-step operation, then HS cannot explain the coda/onset asymmetry described at the outset of this paper. We reexamine this problem and conclude that the data challenge neither the descriptive generalization nor HS.

3 Additional data from Japanese verbs

Kurisu (2012) considers two alternatives, and eventually rejects both. One of these is an analysis based on allomorph selection, which we argue is the correct analysis. Unlike deletion, allomorph selection can be done in a single step, as shown in (4).[2] If allomorphs are both listed as underlying, then choosing either of the allomorphs satisfies faithfulness constraints (Itô and Mester 2004, 2006; Yip 2004; McCarthy 2007b), and as a result, the allomorph is determined by top-ranked CODACOND.

(4) Allomorph selection analysis (Kurisu 2012:318)

/tob-{ru, u}/	CODACOND	MAX_{Root}	$\text{MAX}_{\text{Affix}}$
a. ☞ to.bu			
b. tob.ru	*!		

Kurisu's main argument against allomorph selection is that the allomorphs are phonetically similar, differing only in the presence or absence of the initial coronal. To capture the similarity between these two shapes of the suffixes he posits that coronals must be underlying. This assumes that lexical representations are chosen to maximize phonological predictability, but this assumption does not always hold, as in the well-known case of Maori passive (Hale 1973).[3] As an alternative to shared underlying representations, Itô and Mester (2004) propose a constraint ALLOCORR which prefers identical allomorphs. Sano (2015) further illustrates how sound changes in Japanese are caused by ALLOCORR. In short, the similarity between two allormophs does not need to be explained by way of shared underlying representation: it can (and perhaps should) instead be modelled using violable constraints.

We now provide several kinds of evidence to support the allomorph selection analysis over deletion. Four pieces of data come from Japanese, complemented with further data from other languages. (See Vance 1987:§12 for a comprehensive description of Japanese verb morphology.)

First, not all suffix-initial coronals delete. For example, the past tense suffix /ta/ is never realized as [a] (5). Instead, consonant clusters with [t] of this suffix are resolved by different repairs (coda nasalization, place assimilation, gemination, vowel epenthesis—but never onset deletion). Continuative /te/ behaves the same way.

(5) No coronal deletion of the past tense /ta/

tob + ta	→	tonda	'flied'	Coda nasalization
kam + ta	→	kanda	'bit'	Nasal place assimilation
kaw + ta	→	katta	'bought'	Gemination
kar + ta	→	karita	'borrowed'	Vowel epenthesis
kas + ta	→	kaʃita	'rented'	Vowel epenthesis

Second, verbal compounds preserve coronals and exhibit epenthesis (6). See Poser (1984) for evidence that this vowel is epenthetic. Compare a minimal pair /tob+sase/ → [tob-ase] 'cause to fly' (1) and /tob+sonjiru/ → [tob-i-sonjiru] 'fail to fly'. Both examples involve a

/bs/ cluster, and the purely phonological analysis predicts deletion in both cases.

(6) No coronal deletion in Japanese compounds

tob+dasu	→	tob-i-dasu	'to rush out'
tob+deru	→	tob-i-deru	'to stick out'
tob+sonjiru	→	tob-i-sonjiru	'fail to fly'

Third, not all suffixes differ in terms of the presence/absence of a coronal consonant; Kurisu (2012) only analyzes a subset of the Japanese verbal suffixes. Some pairs do indeed differ in the presence of a coronal consonant (7-a). However, cases in (7-b) seem to involve suppletion, where the two suffixes look entirely different. Some suffixes differ in terms of the presence of a vowel, instead of a consonant (7-c).

(7) a. Pairs that differ in a coronal consonant

	'fly'	'sleep'
Non-past	tob-u	ne-ru
Conditional	tob-eba	ne-reba
Causative	tob-ase	ne-sase
Volitional	tob-oo	ne-joo
Passive	tob-are	ne-rare

b. Pairs do that not fit this generalization

Imperative	tob-e	ne-ro
Potential	tob-e	ne-rare

c. Pairs that differ in the presence of a vowel

Negative	tob-anai	ne-nai
Polite	tob-imasu	ne-masu

Thus, even though some of the coronal suffixes appear to exhibit phonological deletion, upon closer examination this generalization is based on only a subset of the Japanese verbal paradigm patterns. The combined data provides evidence that the different realizations of verbal suffixes are allomorphs, instead of governed by a regular coronal deletion process.

4 Opaque allomorph selection in Japanese

We now move to another type of evidence. The Japanese coronal alternations interact with another process: w-deletion. The interaction is opaque, and hence cannot be dealt with in parallel OT and instead supports an HS analysis. In Japanese, [w] deletes when followed by a non-low vowel, as shown in (8).[4]

(8) W-deletion (Vance 1987; Gibson 2008; Nevins 2011; Tomohiro Yokoyama, p.c.)

		/tob/ 'fly'	/ne/ 'sleep'	/iw/ 'say'	/karakaw/ 'mock'	/maw/ 'dance'	/ow/ 'chase'
Infinitive	/-ru/	tob-u	ne-ru	i~~w~~-u	karaka~~w~~-u	ma~~w~~-u	o~~w~~-u
Subjunctive	/-reba/	tob-eba	ne-reba	i~~w~~-eba	karaka~~w~~-eba	ma~~w~~-eba	o~~w~~-eba
Volitional	/-joo/	tob-oo	ne-joo	i~~w~~-oo	karaka~~w~~-oo	ma~~w~~-oo	o~~w~~-oo
Causative	/-sase/	tob-ase	ne-sase	iw-ase	karakaw-ase	maw-ase	ow-ase

Let us suppose that both w-deletion and coronal deletion satisfy CODACOND. For instance, the input /ow-ru/ 'to chase' could surface as *[o.wu] or *[o.ru]. The former is not well-formed because w-deletion failed to apply before a non-low vowel, while the latter would be the expected output. However, the attested output is [o.u], instead of *[o.ru] with an onset consonant. Why should both [w] and [r] be deleted, leaving an onsetless syllable? An explanation of this puzzle is that w-deletion applies only after allomorph selection. If so, the suffix should be selected first, yielding the intermediate form [ow-u], at which point w-deletion applies, resulting in the correct surface form [o-u]. This is a case of counter-bleeding opacity.

This interaction can be characterized as allomorph selection applying at the intermediate stages of the derivation which is attested cross-linguistically (Gibson 2008; Wolf 2008:§2&3; Nevins 2011:2373−2374). One of the advantages of HS over parallel OT is that it can capture phonological generalizations at intermediate steps.[5]

To capture the Japanese phonotactic restrictions on w+vowel sequences, we propose a markedness constraint *w[−low] (≡ *w* must not be followed by a [−low] vowel). This constraint applies to Japanese phonology in general, only having exceptions in some loanwords (e.g. [witto] 'wit'). The remaining constraints have been used earlier in the squib.

As we have seen above, the allomorph is selected at a step before w-deletion. The selection of the allomorph depends on whether the root ends with a licit coda, which is captured by high ranked CODACOND. When the root ends with a [w], the allomorph without a coronal is selected at the first step (9). Note that when the root ends on an underlying vowel, *-ru* will be selected because of low ranked ONSET.

(9) Step 1: Allomorph selection

/iw-{ru, u}/	CODACOND	*w[−low]	HAVEPLACE	MAX(PLACE)	MAX	ONSET
a. ☞ i.wu		*				
b. iw.ru	*!					

At step 2, the phonotactic constraint *w[−low] drives w-debuccalization (10).

(10) Step 2: Debuccalization

i.wu	CODACOND	*w[−low]	HAVEPLACE	MAX(PLACE)	MAX	ONSET
a. ☞ i.Hu			*	*		
b. i.wu		*!				

At step 3, the placeless onset segment is deleted (11). The derivation converges at step 4.

(11) Step 3: w-deletion

i.Hu	CODACOND	*w[−low]	HAVEPLACE	MAX(PLACE)	MAX	ONSET
a. ☞ i.u					*	*
b. i.Hu			*!			

The Japanese opaque pattern cannot be captured in parallel OT, but can be captured in HS as allomorph selection. The deletion analysis in HS fails, as seen in (3). At step 1, the candidate with a debuccalized coda consonant [iH.ru] would win, ultimately leading to the incorrect winner *[i.ru]. A parallel OT analysis is also unsuccessful, either as deletion (12-a) or allomorphy (b), because the transparent candidate *[i.ru] harmonically bounds the attested opaque output [i.u].

(12) Parallel OT analysis fails

a. Coronal deletion

/iw-ru/	CODACOND	*w[−low]	MAX$_{\text{Root}}$	MAX$_{\text{Affix}}$	ONSET
a. ☹ i.u			*	*!	*!
b. ☞ i.ru			*		
c. i.wu		*!		*	
d. iw.ru	*!				

b. Allomorph selection

/iw-{ru, u}/	CODACOND	*w[−low]	MAX$_{\text{Root}}$	MAX$_{\text{Affix}}$	ONSET
a. ☹ i.u			*		*!
b. ☞ i.ru			*		
c. i.wu		*!			
d. iw.ru	*!				

5 Opaque allomorph selection beyond Japanese

We have demonstrated that once the Japanese data are considered in full detail, the phonological deletion analysis has to be rejected, but the allomorph selection alternative remains viable, and can be successfully captured in HS. Opaque allomorph selection is found in languages other than Japanese, including Ukrainian (Darden 1979; Gibson 2008), German (Kiparsky 1994; Aronoff 1976), Spanish (Aranovich et al. 2005; Aranovich and Orgun 2006), Sanskrit (Kiparsky 1997), Turkish (Lewis 1967; Aranovich et al. 2005; Paster 2006), and Babanki (Akumbu 2015). In this section, we look at allomorph selection in Polish, which interacts opaquely with palatalization. The Polish pattern further illustrates the need for allomorph selection which occurs at an intermediate step of the derivation.

In Polish, the locative suffix *-ɛ* palatalizes root-final alveolars (13-a), but underlyingly palatalized sounds are followed by the *-u* suffix allomorph (b).[6]

(13) Polish masculine nouns (Joanna Chociej, p.c.)

a. Anterior coronals palatalize before front vowels

	‘letter’	‘reptile’	‘dirty man’	‘troublemaker’	‘balloon’
Nominative	list	ɡat	brudas	wɔbus	balɔn
Locative	liɕʨɛ	ɡaʥɛ	brudaɕɛ	wɔbuʑɛ	balɔɲɛ

b. Posterior coronals select the high vowel allomorph instead

	‘leaf’	‘acorn’	‘salmon’	‘type of butterfly’	‘horse’
Nominative	liɕʨ	ʐɔwɛɲʨ	wɔsɔɕ	paɕ	kɔɲ
Locative	liɕʨu	ʐɔwɛɲʥu	wɔsɔɕu	paʑu	kɔɲu

One way of thinking of this pattern is to say that there are two separate operations: *ɛ* → *u* applying after postalveolars, followed by palatalization. If so, this is a case of counter-feeding opacity. Parallel to Japanese, neither of these processes are fully general in the phonology of Polish. For instance, the same opaque alternation is found in the feminine dative case, but the vowel surfaces as a front high vowel *i* (e.g. [kaɕ-ɛ] ‘register-DATIVE’ ∼ [p^{j}ɛrɕ-i] ‘breast-DATIVE’). Furthermore, palatalization applies only at the morpheme boundary before a front vowel.

Now, counter-feeding opacity generally cannot be modeled in HS because that would go against harmonic improvement, that is, a derivation would have to terminate at the step before convergence (McCarthy 2000, 2007a; Elfner 2009). In this particular case, however, HS can model the pattern, but only because we are dealing with allomorph selection. This is shown in tableaux (14)–(15). We use similar markedness constraints as the previous analyses (e.g. Rubach 2003; Łubowicz 2014), except that we choose designations that relate them more straightforwardly to the data at hand (e.g. *ʨɛ instead of the PALATALIZATION constraint). A root that ends on an anterior coronal selects a front allomorph at step 1, under the pressure of top-ranked *tu (14-a-i). At the next step, palatalization applies (b-i), followed by convergence (c-i). (Palatalization should apply in two steps, one for each coronal, but we are leaving out that intermediate step for brevity.) Crucially, raising the vowel would violate IDENT(high), which is not violated by allomorph selection at step 1.

(14) 'Letter': /list-{ɛ, u}/ → listɛ → [liɕʨɛ]

a. Step 1: Allomorph selection

/list-{ɛ, u}/	IDENT(high)	*tu	*tɛ	*ʨɛ
i. ☞ list-ɛ			*	
ii. list-u		*!		

b. Step 2: Palatalization

list-ɛ	IDENT(high)	*tu	*tɛ	*ʨɛ
i. ☞ liɕʨ-ɛ				*
ii. list-ɛ			*!	

c. Step 3: Convergence

liɕʨ-ɛ	IDENT(high)	*tu	*tɛ	*ʨɛ
i. ☞ liɕʨ-ɛ				*
ii. liɕʨ-u	*!			

Roots with underlying posterior coronals instead select the *u*-allomorph (15-a-i), followed by convergence (15-b-i). Again, IDENT(high) plays a crucial role.

(15) ‘Leaf’: /liɕt͡ɕ-{ɛ, u}/ → [liɕt͡ɕu]

a. Step 1: Allomorph selection

/liɕt͡ɕ-{e, u}/	IDENT(high)	*tu	*tɛ	*t͡ɕɛ
i. ☞ liɕt͡ɕ-u				
ii. liɕt͡ɕ-ɛ				*!

b. Step 2: Convergence

liɕt͡ɕ-u	IDENT(high)	*tu	*tɛ	*t͡ɕɛ
i. ☞ liɕt͡ɕ-u				
ii. liɕt͡ɕ-ɛ	*!			*

Similar to the Japanese case in (12), Polish cannot be captured in parallel OT. Both kinds of opaque interactions can be captured as allomorph selection in HS.

6 Conclusions

Kurisu (2012) argues that Japanese has onset deletion that cannot be analyzed as a two-step process in Harmonic Serialism. This challenges an otherwise robust cross-linguistic generalization that the second consonant never deletes in cluster simplification. In this squib, we have shown that Japanese does not involve deletion but rather allomorph selection. This conclusion is corroborated by reexamination of the data and opaque interactions with a phonological pattern. Furthermore, we have presented another case of opaque allomorph selection in Polish, which shows that the Japanese pattern is not a cross-linguistically isolated example. The reevaluation of the data thus removes a challenge to two-step deletion in Harmonic Serialism and further shows that HS can capture many cases of opaque allomorphy, which parallel OT cannot. This paper also shows that the descriptive generalization—that onset consonants never delete in cluster simplification—remains robust, and that any successful theory of phonology needs to capture this generalization.

References

Akumbu, Pius W. 2015. Single URs vs. allomorphy: The case of Babanki coda consonant deletion. In *UC Berkeley Phonology Lab annual report*, ed. Jevon Heath, 191–209. Berkeley: University of California, Berkeley.

Aranovich, Raúl, Sharon Inkelas, Orhan Orgun, and Ronald Sprouse. 2005. Opacity in phonologically conditioned suppletion. Paper presented at 13th Manchester Phonology Meeting.

Aranovich, Raúl, and Orhan Orgun. 2006. Opacity in *-ez/-eza* suffixation. In *Selected proceedings of the 8th Hispanic Linguistics Colloquium*, ed. Timothy L. Face and Carol A. Klee, 116–122. Somerville, MA: Cascadilla Press.

Aronoff, Mark. 1976. *Word formation in Generative Grammar*. Cambridge, MA: MIT Press.

Beckman, Jill N. 1998. Positional Faithfulness. Doctoral Dissertation, University of Massachusetts, Amherst. Available on Rutgers Optimality Archive, ROA 234, http://roa.rutgers.edu.

Darden, Bill. 1979. On the nature of morphophonemic rules. In *Papers from the Fifteenth Regional Meeting, Chicago Linguistic Society*, ed. Paul R. Clyne, William F. Hanks, and Carol L. Hofbauer, 79–89. Chicago: Chicago Linguistic Society.

Elfner, Emily. 2009. Syllabification and stress-epenthesis interactions in Harmonic Serialism. Ms. University of Massachussets, Amherst. 50pp.

Gibson, Masayuki. 2008. Opaque allomorph selection: The need for intermediate forms. Paper presented at the 82th Annual Meeting of the Linguistic Society of America (Chicago, IL).

Goldsmith, John A. 1990. *Autosegmental and metrical phonology*. Oxford: Blackwell.

Hale, Kenneth. 1973. Deep-surface canonical disparities in relation to analysis and change: an Australian example. In *Diachronic, areal, and typological linguistics (Current trends in linguistics 11)*, ed. Thomas A. Sebeok, 401–458. The Hague: Mouton.

Itô, Junko. 1986/1988. *Syllable theory in prosodic phonology*. New York: Garland.

Itô, Junko. 1989. A prosodic theory of epenthesis. *Natural Language and Linguistic Theory* 7:217–259.

Itô, Junko, and Armin Mester. 1998. Markedness and word structure: OCP effects in Japanese. Ms. Available on Rutgers Optimality Archive, ROA 255, http://roa.rutgers.edu.

Itô, Junko, and Armin Mester. 2003. On the sources of opacity in OT: coda processes in German. In *The Syllable in Optimality Theory*, ed. Caroline Féry and Ruben van de Vijver, 271–303. Cambridge: Cambridge University Press.

Itô, Junko, and Armin Mester. 2004. Morphological contrast and merger: ranuki in Japanese. *Journal of Japanese Linguistics* 20:1–18.

Itô, Junko, and Armin Mester. 2006. Indulgentia parentum filiōrum perniciēs: Lexical allomorphy in Latin and Japanese. In *Wondering at the natural fecundity of things: Essays in honor of Alan Prince*, 185–194. Santa Cruz, CA: Linguistics Research Center.

Jesney, Karen. 2011. Positional faithfulness, non-locality, and the Harmonic Serialism solution. In *Proceedings of NELS 39*, ed. Susi Lima, Kevin Mullin, and Brian Smith, 429–440. Amherst: GLSA, University of Massachusetts.

Kimper, Wendell. 2011. Locality and globality in phonological variation. *Natural Language and Linguistic Theory* 29:423–465.

Kiparsky, Paul. 1994. Allomorphy or morphophonology? In *Trubetzkoy's Orphan: Proceedings of the Montréal Roundtable "Morphonology: Contemporary responses"*, ed. Rajendra Singh and Richard Desroches, 13–31. Amsterdam: John Benjamins.

Kiparsky, Paul. 1997. LP and OT. Course handout from LSA Summer Institute, Cornell University, Ithaca, NY.

Kurisu, Kazutaka. 2012. Fell-swoop onset deletion. *Linguistic Inquiry* 43:309–321.

Lewis, G. L. 1967. *Turkish grammar*. Oxford: Oxford University Press.

Łubowicz, Anna. 2006. Opaque allomorphy in OT. In *Proceedings of the 25rd West Coast Conference on Formal Linguistics*, ed. Donald Baumer, David Montero, and Michael Scanlon. Somerville, MA: Cascadilla Press.

Łubowicz, Anna. 2007. Paradigmatic contrast in Polish. *Journal of Slavic Linguistics* 15:229–262.

Łubowicz, Anna. 2012. *The phonology of contrast*. London: Equinox.

Łubowicz, Anna. 2014. Contrast preservation in Polish palatalization. Paper presented at the Palatalization Conference. University of Tromsø, December 4.

McCarthy, John J. 2000. Harmonic Serialism and parallelism. In *Linguistics department faculty publication series*, 1–25. Amherst, MA: University of Massachusetts.

McCarthy, John J. 2007a. *Hidden generalizations: Phonological opacity in Optimality Theory*. London: Equinox Publishing Company.

McCarthy, John J. 2007b. Slouching towards optimality: Coda reduction in OT-CC. *Phonological Studies (Journal of the Phonological Society of Japan)* 7:89–104.

McCarthy, John J. 2008. The gradual path to cluster simplification. *Phonology* 25:271–319.

McCarthy, John J. 2010a. Harmonic Serialism supplement to *Doing Optimality Theory*. Ms. University of Massachussets Amherst. Available on Rutgers Optimality Archive, ROA 1099, http://roa.rutgers.edu.

McCarthy, John J. 2010b. An introduction to Harmonic Serialism. *Language and Linguistics Compass* 10:1010–1018.

McCarthy, John J. to appear. The theory and practice of Harmonic Serialism. In *Harmonic Grammar and Harmonic Serialism*, ed. John J. McCarthy and Joe Pater. London: Equinox.

McCarthy, John J., and Alan Prince. 1995. Faithfulness and reduplicative identity. In *University of Massachusetts occasional papers in linguistics 18: Papers in Optimality Theory*, ed. Jill N. Beckman, Laura Walsh, and Suzanne Urbanczyk, 249–384. Amherst: GLSA, University of Massachusetts. Available on Rutgers Optimality Archive, ROA 60, http://roa.rutgers.edu.

Nevins, Andrew. 2011. Phonologically conditioned allomorph selection. In *The Blackwell companion to phonology*, ed. Marc van Oostendorp, Colin J. Ewen, Elizabeth Hume, and Keren D. Rice, 2357–2382. Malden, MA: Blackwell.

Paster, Mary. 2006. Phonological conditions on affixation. Doctoral Dissertation, University of California Berkeley.

Poser, William J. 1984. The phonetics and phonology of tone and intonation in Japanese. Doctoral Dissertation, Massachusetts Institute of Technology, Cambridge.

Pruitt, Kathryn. 2010. Serialism and locality in constraint-based metrical parsing. *Phonology* 27:481–526.

Pruitt, Kathryn. 2012. Stress in Harmonic Serialism. Doctoral Dissertation, University of Massachusetts, Amherst.

Rubach, Jerzy. 2003. Polish palatalization in derivational optimality theory. *Lingua* 113:197–237.

Sanders, Nathan. 2003. Opacity and sound change in the Polish lexicon. Doctoral Dissertation, University of California, Santa Cruz. Available on Rutgers Optimality Archive, ROA 603, http://roa.rutgers.edu.

Sano, Shin-ichiro. 2015. Optimization of the verbal inflectional paradigm by the cyclic application of morphophonological processes: Evidence from potential forms in Japanese. *Open Linguistics* 1:580–595.

Steriade, Donca. 2001. Directional asymmetries in place assimilation: a perceptual account. In *The role of speech perception in phonology*, ed. Elizabeth V. Hume and Keith Johnson, 219–250. New York: Academic Press.

Torres-Tamarit, Francesc, and Peter Jurgec. 2015. Lapsed derivations: Ternary stress in Harmonic Serialism. *Linguistic Inquiry* 46:376–387.

Vance, Timothy J. 1987. *An introduction to Japanese phonology*. Albany, NY: State University of New York Press.

Wilson, Colin. 2000. Targeted constraints: an approach to contextual neutralization in Optimality Theory. Doctoral Dissertation, Johns Hopkins University, Baltimore, MD.

Wolf, Matthew Adam. 2008. Optimal Interleaving: Serial phonology-morphology interaction in a constraint-based model. Doctoral Dissertation, University of Massachusetts, Amherst. Available on Rutgers Optimality Archive, ROA 996, http://roa.rutgers.edu.

Yip, Moira. 2004. Phonological markedness and allomorph selection in Zahao. *Language and Linguistics* 5:969–1001.

Notes

[1]McCarthy (2007b) recognizes this as a potential example, suggesting that the pattern could be epenthetic. The epenthesis analysis does not explain why different types of coronal consonants can occur suffix-initially.

[2]For a two-step analysis, see Wolf (2008). According to this alternative, Japanese roots are spelled out first, and their shape determines which suffix allomorph is selected.

[3]Thanks to an anonymous reviewer for pointing this out. Another case can be found in Babanki where /ŋ/ appears to be deleted. Under closer examination the alternation is revealed to involve allomorph selection (Akumbu 2015).

[4]Note that this alternation is a case of deletion rather than epenthesis. An anonymous reviewer asks whether [w] in words like [maw-ase] 'to cause to dance' or [maw-anai] 'not dance' can be epenthetic. We could postulate that in verbal paradigms, [w] is inserted between two vowels. However, this epenthesis alternative fails to explain how vowel-final stems behave when followed by vowel-initial suffixes. For instance, the negative form of the root /ne/ 'to sleep' is [nenai], instead of *[newanai] (7-c). If [w] were epenthetic, rather than being a part of the stem as we posit, why would this form not arise? Another reason to assume that this /w/ is a part of the stem is because /w/ causes gemination in the past tense: /kaw+ta/ $\rightarrow$ [katta] 'bought' (5). Vowel-final stems do not undergo this gemination, [neta] 'slept'. It is highly unlikely that [w] is inserted to be geminated.

[5]For instance, footing sometimes ignores subsequent syncope, which can only be captured by grammars that can assign footing before applying syncope, such as HS (McCarthy 2010b). Other advantages of HS include predictions about variation (Kimper 2011), positional faithfulness (Jesney 2011), and stress typology (Pruitt 2010, 2012; Torres-Tamarit and Jurgec 2015). An anonymous reviewer asks whether other multi-level versions of Optimality Theory can account for the data in question. Since Kurisu (2012) focuses on the HS vs. Parallel OT debate, we do not go into detail about other multi-level models. We note, however, that only HS can account for the onset/coda asymmetry discussed at the outset of the paper.

[6]For previous treatments of this kind of opaque allomorph selection, see Rubach (2003); Sanders (2003); Łubowicz (2006, 2007, 2012, 2014).